Integration

Questions 1–101

1.36 — Question 1. Find

a) $\int (8x+1)\,dx$

b) $\int (6x^2-3)\,dx$

c) $\int (10x^3+9)\,dx$

d) $\int (3x+5)(4x^2-1)\,dx$

1.36 — Question 2. Find

a) $\int \left(\frac{2}{3}x+\frac{5}{3}x^4\right)\,dx$

b) $\int \left(\frac{3}{4}x^2-\frac{14}{3}x^6\right)\,dx$

c) $\int \left(\frac{2}{5}x^7-\frac{3}{8}x^5\right)\,dx$

d) $\int \left(\frac{1}{3}x^2+4\right)\left(\frac{2}{3}x^2+6\right)\,dx$

1.36 — Question 3. The curve y has gradient function $\frac{dy}{dx} = 2 - 2x$. Sketch the family of solutions that solve this differential equation.

1.36 — Question 4. The curve y has gradient function $\frac{dy}{dx} = \left(\frac{x}{2} - 1\right)\left(\frac{x}{3} + 1\right)$ and passes through the point $(6, 1)$. Show that y can be written in the form $y = ax^3 + bx^2 + cx + d$, where a, b, c and d are rational numbers.

1.36 — Question 5. Find the equation of the curve y given the gradient function is:

a) $\dfrac{dy}{dx} = x^{2/3}$

b) $\dfrac{dy}{dx} = 6x^{1/5}$

c) $\dfrac{dy}{dx} = 21x^{3/4}$

d) $\dfrac{dy}{dx} = (2x^{1/6} + 3)^2$

1.36 — Question 6. Find

a) $\displaystyle\int (x^3 + x^{3/2})\,dx$

b) $\displaystyle\int (5x^{-2} - 2x^{2/3})\,dx$

c) $\displaystyle\int (8x^{-3} - 4x^{6/5})\,dx$

d) $\displaystyle\int (2x^{-4} - 3x^{4/3})^2\,dx$

1.36 — Question 7. Find $\int (1 - x^{-2})(1 - x^{-4})\, dx$

1.36 — Question 8. The curve y has gradient function $\dfrac{dy}{dx} = 40x^{1/3} - 4x$ and passes through the point $(1, -1)$. Show that y can be written in the form $y = ax^{4/3} + bx^2 + c$, where a, b and c are integers.

1.36 — Question 9. Find the equation of the curve y given the gradient function is:

a) $\dfrac{dy}{dx} = 5 - 2x^{-4}$

b) $\dfrac{dy}{dx} = 6x^{-3}(x - 2)$

c) $\dfrac{dy}{dx} = x^{1/2}(8x - 3)$

d) $\dfrac{dy}{dx} = (2x^{3/2} - 3)(2x^{-7/2} + 7)$

1.36 — Question 10. Find

a) $\displaystyle\int 4\sqrt{x}(x - 1)\,dx$

b) $\displaystyle\int \sqrt[4]{x}\,dx$

c) $\displaystyle\int \sqrt[5]{32x}\,dx$

d) $\displaystyle\int \sqrt[4]{81x^{1/4}}\,dx$

1.36 — Question 11. The curve y has gradient function $\dfrac{dy}{dx} = x^2(7\sqrt{x} - 3)$ and passes through the point $(1, -4)$. Show that y can be written in the form $y = ax^{7/2} - bx^3 + c$, where a, b and c are integers.

1.36 — Question 12. Find the equation of the curve y given the gradient function is:

a) $\dfrac{dy}{dx} = \dfrac{4}{x^2}$

b) $\dfrac{dy}{dx} = \dfrac{6}{x^3}$

c) $\dfrac{dy}{dx} = \dfrac{9}{5x^4}$

d) $\dfrac{dy}{dx} = \dfrac{100}{9x^{10}}$

1.36 — Question 13. Find

a) $\int \frac{4}{3\sqrt{x}}\,dx$

b) $\int \frac{6}{\sqrt[3]{x}}\,dx$

c) $\int \left(\frac{8}{\sqrt[5]{x}} - \frac{100}{\sqrt[3]{8x}}\right)\,dx$

d) $\int \left(\frac{1}{\sqrt[3]{x}} - \frac{3}{x\sqrt{x}}\right)^2\,dx$

1.36 — Question 14. The curve y has gradient function $\frac{dy}{dx} = \frac{8}{\sqrt{x}}$ and passes through the point $(16, -8)$. Show that y can be written in the form $y = a(2\sqrt{x} + b)$, where a and b are integers.

1.36 — Question 15. The curve y has gradient function $\frac{dy}{dx} = \frac{750}{x^4}$ and passes through the point $(5, 248)$. Show that y can be written in the form $y = a\left(1 - \frac{1}{x^3}\right)$, where a is an integer.

1.36 — Question 16. The curve y has gradient function $\frac{dy}{dx} = \frac{6}{x\sqrt{x}}$ and passes through the point $(8, 0)$. Show that y can be written in the form $y = a\left(\sqrt{b} + \frac{c}{\sqrt{x}}\right)$, where a, b and c are integers.

1.36 — Question 17. The curve y has gradient function $\frac{dy}{dx} = \frac{12}{\sqrt[3]{x}}$ and passes through the point $(27, 0)$. Show that y can be written in the form $y = a(x^{2/3} + b)$, where a and b are integers.

1.36 — Question 18. The curve y has gradient function $\frac{dy}{dx} = \frac{1}{x^3\sqrt{x}}$ and passes through the point $\left(1, \frac{3}{5}\right)$. Show that y can be written in the form $y = a - \frac{b}{cx^2\sqrt{x}}$, where a, b and c are integers.

1.36 — Question 19. Find $\int \left(\frac{4}{x^2\sqrt{x}} - \frac{\sqrt{x}}{4x^2} \right) dx$

1.36 — Question 20. Find $\int \left(\frac{6}{5x^3\sqrt{x}} - \frac{\sqrt[3]{x}}{6x^3} \right) dx$

1.36 — Question 21. Find $\int \left(\frac{3\sqrt{x}}{4x^4} - \frac{x\sqrt{x}}{\sqrt[3]{x}} \right) dx$

1.36 — Question 22. It is given that $f'(x) = \left(3x - \frac{4}{x}\right)^2$ and $f(2) = -2$. Show that $f(x) = ax^3 + bx + \frac{c}{x} + d$ where a, b, c and d are integers.

1.36 — Question 23. It is given that $k > 0$. Find the equation of the curve y given the gradient function is:

a) $\dfrac{dy}{dx} = kx^4$

b) $\dfrac{dy}{dx} = (kx)^4$

c) $\dfrac{dy}{dx} = \dfrac{1}{kx^4}$

d) $\dfrac{dy}{dx} = \dfrac{18}{(2kx)^5}$

1.36 — Question 24. Find

a) $\displaystyle\int \frac{5 + x^3}{x^2}\, dx$

b) $\displaystyle\int \frac{3x - 2x^3}{x^5}\, dx$

c) $\displaystyle\int \frac{8x^2 - 9}{x^4}\, dx$

d) $\displaystyle\int \frac{30x^3 - 7}{14x^5}\, dx$

1.36 — Question 25. Find the equation of the curve y given the gradient function is:

a) $\dfrac{dy}{dx} = \dfrac{5x^4 - 7}{5x^3}$

b) $\dfrac{dy}{dx} = \dfrac{3 - 8x^3}{4x^5}$

c) $\dfrac{dy}{dx} = \dfrac{6x^7 - 8}{12x^4}$

d) $\dfrac{dy}{dx} = \dfrac{8x - 3x^2 - 6}{24x^5}$

1.36 — Question 26. It is given that $f'(x) = \dfrac{x - 8}{x^3}$ and $f(1) = 0$. Show that $f(x) = \dfrac{ax^2 + bx + c}{x^2}$ where a, b and c are integers.

1.36 — Question 27. It is given that $f'(x) = \left(\dfrac{14-7x}{\sqrt[3]{x}}\right)^2$ and $f(1) = 462$. Show that $f(x) = 21x^{1/3}(ax^2+bx+c)$ where a, b and c are integers.

1.36 — Question 28. Prerequisite knowledge for Q28: Stationary Points

A curve has gradient function $\dfrac{dy}{dx} = 3x^2 + kx - 21$. The curve has two turning points. One of the turning points is at $(1, -6)$. Find the coordinates of the other turning point.

1.36 — Question 29. Prerequisite knowledge for Q29–Q30: The Factor Theorem
A curve has gradient function $f'(x) = kx^2 + 74x + 11$ where $f(-1) = -100$. It is also given that $(x + 5)$ is a factor of $f(x)$. Find $f(x)$ in its simplest form.

1.36 — Question 30. A curve has gradient function $f'(x) = 576x^3 + ax^2 - 208x + 40$. It is given that $(6x - 5)$ and $(2x + 1)$ are factors of $f(x)$. Find $f(x)$ in its simplest form.

1.36 — Question 31. Prerequisite knowledge for Q31: Binomial Expansion
Find

a) $\int [(5+x)^4 - (5-x)^4]\,dx$

b) $\int [(3+2x)^5 - (3-2x)^5]\,dx$

c) $\int [(1-3x)^6 - (1+3x)^6]\,dx$

d) $\int [(1+2x)^7 - (1-2x)^7]\,dx$

1.36 — Question 32. The curve y has gradient function $\frac{dy}{dx} = (4+2x)^3 - (4-2x)^3$ and passes through the point $(-2, 1024)$. Show that y can be written in the form $y = 4(x^2 + a)^2$, where a is an integer.

1.36 — Question 33. Prerequisite knowledge for Q33: Second Derivatives
Find the equation of the curve y given:

a) $\frac{d^2y}{dx^2} = 3$

b) $\frac{d^2y}{dx^2} = 4x - 5$

c) $\frac{d^2y}{dx^2} = 6x^2 - 8x + 9$

d) $\frac{d^2y}{dx^2} = 8x^3 - 7x^5$

1.36 — Question 34. Find the equation of the curve y given:

a) $\frac{d^2y}{dx^2} = \sqrt{x}$

b) $\frac{d^2y}{dx^2} = \frac{5}{\sqrt{x}}$

c) $\frac{d^2y}{dx^2} = \frac{3}{x^3} - \frac{x^3}{3}$

d) $\frac{d^2y}{dx^2} = \frac{8}{x^4\sqrt{x}}$

1.36 — Question 35. Find the equation of the curve y given $\dfrac{d^3y}{dx^3} = \dfrac{10}{x\sqrt{x}}$.

1.36 — Question 36. It is given that the curve $y = f(x)$, where $f(-3) = 2$ and $f'(3) = 2$. It is also given that $f''(x) = 14 - 6x$. Find the equation of the curve $y = f(x)$ in its simplest form.

1.36 — Question 37. It is given that the curve $y = f(x)$, where $f(1) = 3$ and $f'(1) = 5$. It is also given that $f''(x) = 36x^2 + 24x + 12$. Find the equation of the curve $y = f(x)$ in its simplest form.

1.36 — Question 38. It is given that the curve $y = f(x)$, where $f(2) = 5$ and $f'(1) = -2$. It is also given that $f''(x) = \dfrac{18}{x^3}$. Find the equation of the curve $y = f(x)$ in its simplest form.

1.36 — Question 39. It is given that the curve $y = f(x)$, where $f(1) = 11$ and $f'(0) = 0$. It is also given that $f''(x) = 210(x - \sqrt{x} + x\sqrt{x})$. Find the equation of the curve $y = f(x)$ in its simplest form.

1.36 — Question 40. Prerequisite knowledge for Q40: Stationary Points

It is given that the curve $y = f(x)$, where $f(2) = -37$ and $f'(1) = -15$. It is also given that $f''(x) = 12(kx - 1)$ and the curve has a stationary point at $x = \frac{1}{2}$. Find the equation of the curve $y = f(x)$ in its simplest form.

1.37 — Question 1. Evaluate the following definite integrals:

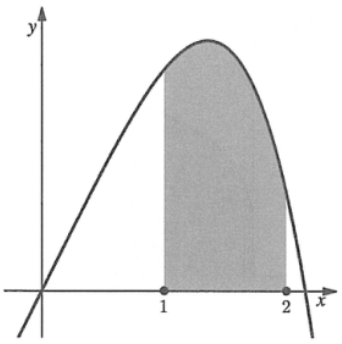
a) $\int_2^6 (8x+1)\,dx$

b) $\int_3^4 (6x^2-1)\,dx$

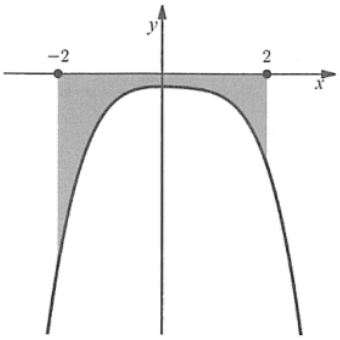
c) $\int_4^8 (500+4x-x^3)\,dx$

d) $\int_{-1}^1 (14x-9x^5)\,dx$

1.37 — Question 2. The diagram below shows the curve with equation $y = 10x - x^4$. Find the exact value of the shaded area.

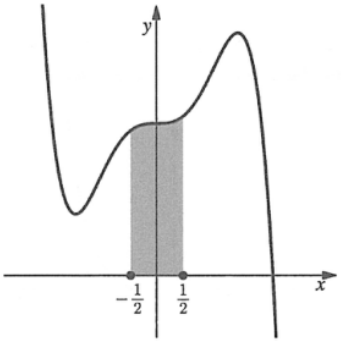


1.37 — Question 3. The diagram below shows the curve with equation $y = -2 - x^2 + x^3 - x^4$. Find the exact value of the shaded area.



1.37 — Question 4. It is given that $\int_1^3 (f(x) + 2)\,dx = 50$. Find the value of $\int_1^3 f(x)\,dx$.

1.37 — Question 5. The diagram below shows the curve with equation $y = 10 - x^5 + 4x^3$. Find the exact value of the shaded area.

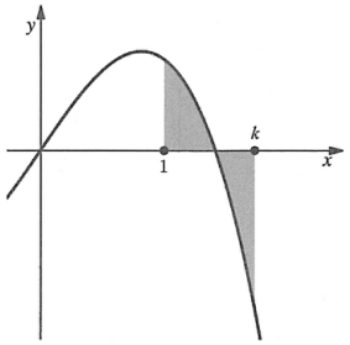


1.37 — Question 6.

a) Sketch the graph of $y = (x - 1)(4 - x)$.

b) Explain why $\int_1^5 (x - 1)(4 - x)\,dx$ does not give the area between the curve, the x-axis and the lines $x = 1$ and $x = 5$.

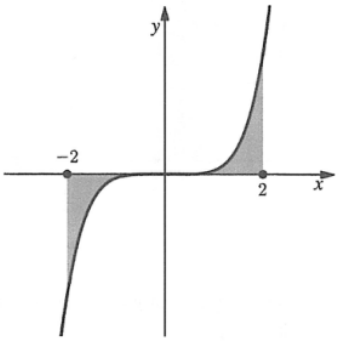
1.37 — Question 7. The diagram below shows the curve with equation $y = 4x - 2x^3$. It is given that $\int_1^k (4x - 2x^3)\,dx = 0$ where k is a positive constant and $k > 1$. Find the value of k.



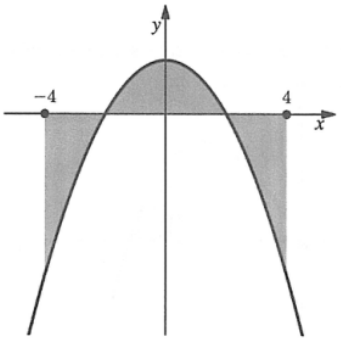
1.37 — Question 8. A quadratic curve has the equation $y = ax^2 + bx + c$. It is given that the curve passes through $(-2, 0)$, $(6, 0)$ and $(0, 240)$. Find the exact finite area between the quadratic curve and the x-axis.

1.37 — Question 9. Find the value of the constant k given that $\int_{-3}^{3} (kx^2 - 5x^4)\,dx = 18$.

1.37 — Question 10. The diagram below shows the curve with equation $y = x^5$. Find the exact value of the shaded area.



1.37 — Question 11. The diagram below shows the curve with equation $y = 4 - x^2$. Find the exact value of the shaded area.



1.37 — Question 12. Show that $\int_{\sqrt{2}}^{\sqrt{3}} (3x^2 + 1)\, dx = 4\sqrt{3} - 3\sqrt{2}$.

1.37 — Question 13. Show that $\int_{1}^{1+\sqrt{2}} (50 - 6x^2)\,dx = 40\sqrt{2} - 12$.

1.37 — Question 14. Show that $\int_{1+\sqrt{3}}^{2+\sqrt{3}} 3x(x+2)\,dx = 25 + 15\sqrt{3}$.

1.37 — Question 15. Find the exact value of the following definite integrals:

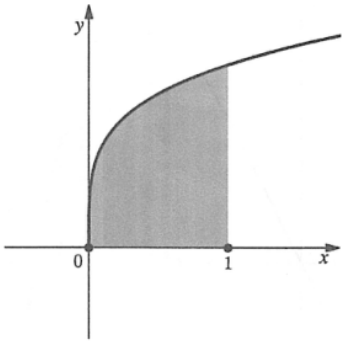
a) $\displaystyle\int_4^{16} \sqrt{x}\,dx$

b) $\displaystyle\int_{-2}^{-1} \frac{6}{x^2}\,dx$

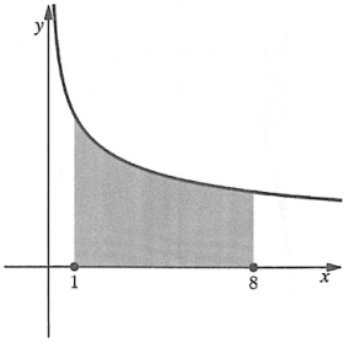
c) $\displaystyle\int_1^{9} \frac{4}{\sqrt{x}}\,dx$

d) $\displaystyle\int_1^{64} \left(\frac{1}{\sqrt[3]{x}} - \frac{1}{\sqrt[4]{x}}\right) dx$

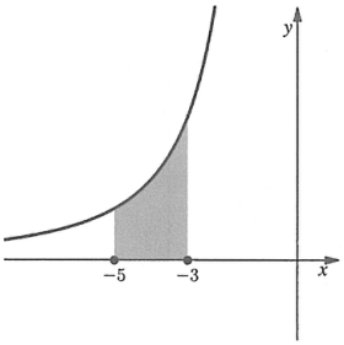
1.37 — Question 16. The diagram below shows the curve with equation $y = \sqrt[4]{x}$. Find the exact value of the shaded area.



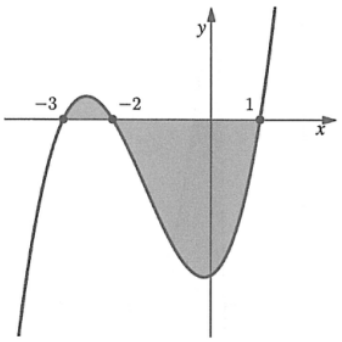
1.37 — Question 17. The diagram below shows the curve with equation $y = \dfrac{12}{\sqrt[3]{x}}$. Find the exact value of the shaded area.



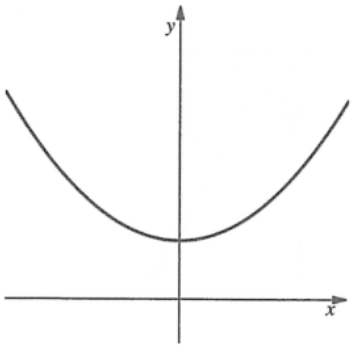
1.37 — Question 18. The diagram below shows the curve with equation $y = \dfrac{9}{x^2}$. Find the exact value of the shaded area.



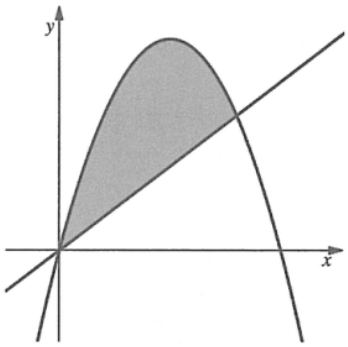
1.37 — Question 19. The diagram below shows the curve with equation $y = x^3 + 4x^2 + x - 6$. Find the exact value of the shaded area.



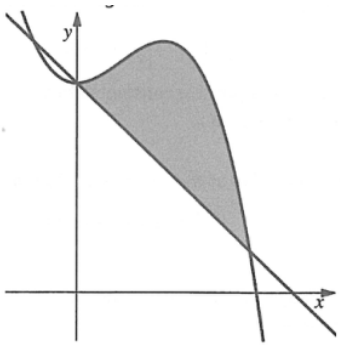
1.37 — Question 20. The diagram below shows the curve with equation $y = \frac{1}{2}x^2 + 3$. Find the area bounded by the curve, the x-axis, and the lines $x = k$ and $x = 2k$, giving your answer in terms of k, where k is a non-zero constant.



1.37 — Question 21. The diagram below shows the curve with equation $y = x(5 - x)$ and the line with equation $y = x$. Find the exact value of the shaded region.



1.37 — Question 22. The diagram below shows the curve with equation $y = 20 - x^2(x - 3)$ and the line with equation $y = 20 - 4x$. Find the exact value of the shaded region.



1.37 — Question 23. Find the value of the constant k given that $\int_1^2 \left(\frac{k}{x^3} + 2\right) dx = 5$.

1.37 — Question 24. Find the value of the constant k given that $\int_1^9 (2kx - 3\sqrt{x})\, dx = 108$.

1.37 — Question 25. Find the possible values of the constant k given that $\int_1^{16} (k\sqrt{x} + 2x + 2)\,dx = 3k^2$.

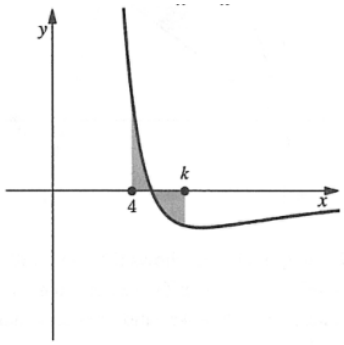
1.37 — Question 26. Find the exact finite area enclosed between the curve $y = 6x^2 - 5x + 1$ and the x-axis.

1.37 — Question 27. Find the maximum value of $\int_2^4 (3p - 2p^2x)\,dx$ where p is a positive constant.

1.37 — Question 28. The diagram below shows the curve with equation $y = \frac{5}{x^3} - \frac{1}{x^2}$. It is given that

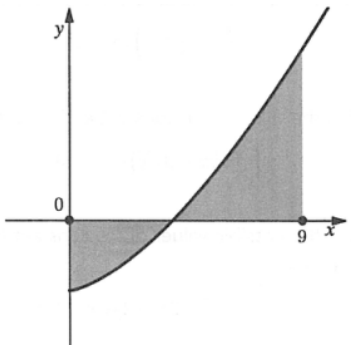
$$\int_4^k \left(\frac{5}{x^3} - \frac{1}{x^2}\right) dx = 0$$

where k is a positive constant with $k > 4$. Find the value of k.

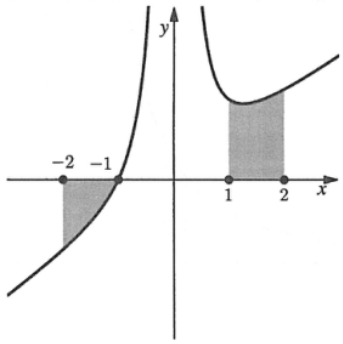


1.37 — Question 29. Find the value of the constant k given that $\int_1^k \left(\frac{30}{\sqrt{x}} + 10\right) dx = 6090$.

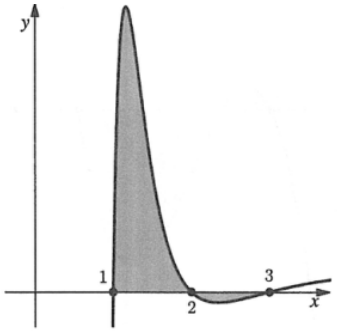
1.37 — Question 30. The diagram below shows the curve with equation $y = x\sqrt{x} - 8$. Find the exact value of the shaded area.



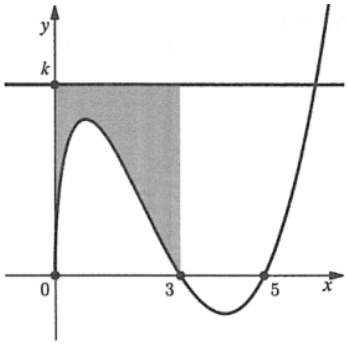
1.37 — Question 31. The diagram below shows the curve with equation $y = x + \frac{1}{x^2}$. Find the exact value of the shaded area.



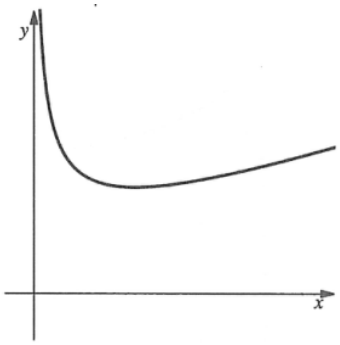
1.37 — Question 32. The diagram below shows the curve with equation $y = \frac{1}{x^2} - \frac{6}{x^3} + \frac{11}{x^4} - \frac{6}{x^5}$. Find the exact value of the shaded area.



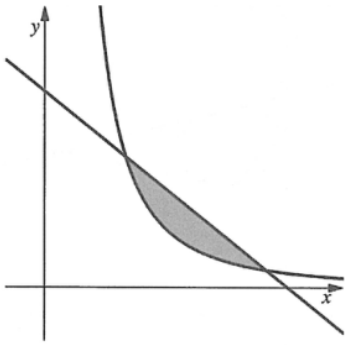
1.37 — Question 33. The diagram below shows the curve with equation $y = 35(x-3)(x-5)\sqrt{x}$ and the line $y = k$. The shaded area is $300\sqrt{3}$ units^2. Find the value of k.



1.37 — Question 34. The diagram below shows the curve with equation $y = \dfrac{2}{\sqrt{x}} + x$. Find the area bounded by the curve, the x-axis, and the lines $x = k$ and $x = 9k$, giving your answer in terms of k, where k is a positive constant.



1.37 — Question 35. The diagram below shows the curve with equation $y = \dfrac{2}{x^2}$ and the line with equation $y = 3 - x$. Find the exact value of the shaded region.



1.37 — Question 36. Find the exact value of the following definite integrals:

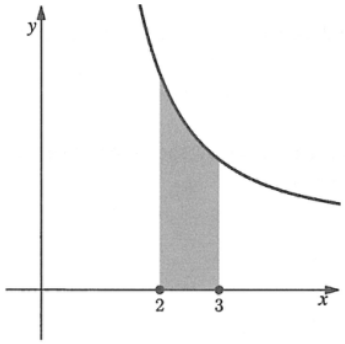
a) $\displaystyle\int_1^3 \frac{x+2}{x^3}\,dx$

b) $\displaystyle\int_2^4 \frac{3x+8}{x^4}\,dx$

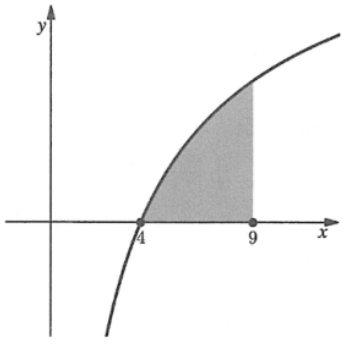
c) $\displaystyle\int_{-3}^{-1} \frac{4-5x}{2x^3}\,dx$

d) $\displaystyle\int_{-4}^{-2} \frac{3x^2-2}{6x^4}\,dx$

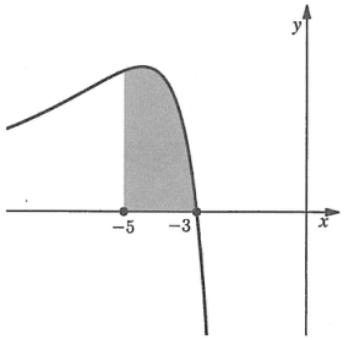
1.37 — Question 37. The diagram below shows the curve with equation $y = \dfrac{x^2 + 10}{x^2}$. Find the exact value of the shaded area.



1.37 — Question 38. The diagram below shows the curve with equation $y = \dfrac{\sqrt{x} - 2}{\sqrt{x}}$. Find the exact value of the shaded area.



1.37 — Question 39. The diagram below shows the curve with equation $y = \dfrac{x+3}{x^3}$. Find the exact value of the shaded area.



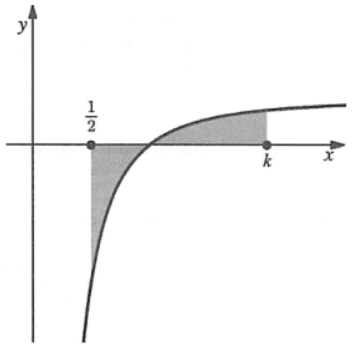
1.37 — Question 40. Find the value of the constant k given that $\displaystyle\int_1^2 \frac{kx+1}{x^4}\,dx = \frac{11}{12}$.

1.37 — Question 41. Find the value of the constant k given that $\displaystyle\int_4^9 \frac{4-k\sqrt{x}}{6\sqrt{x}}\,dx = 1$.

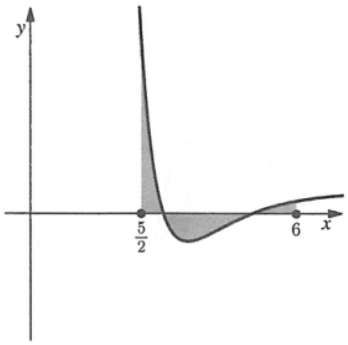
1.37 — Question 42. The diagram below shows the curve with equation $y = \dfrac{x^2-1}{x^2}$. It is given that

$$\int_{1/2}^{k} \frac{x^2-1}{x^2}\,dx = 0$$

where k is a positive constant and $k > \dfrac{1}{2}$. Find the value of k.

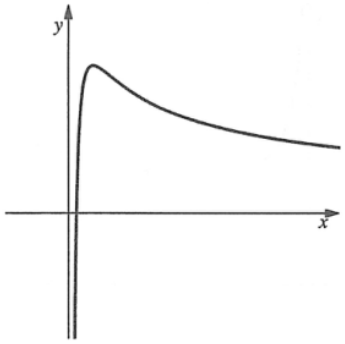


1.37 — Question 43. The diagram below shows the curve with equation $y = \dfrac{(3-x)(5-x)}{x^4}$. Find the exact value of the shaded area.

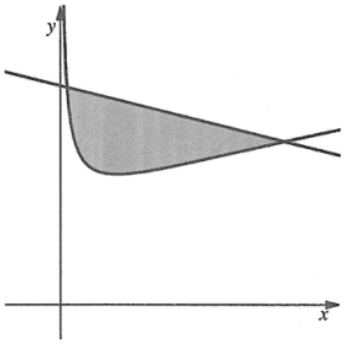


1.37 — Question 44. Find the exact area enclosed between the curve $y = 84x^{5/2}$, the x-axis, and the line $x = 2$.

1.37 — Question 45. The diagram below shows the curve with equation $y = \dfrac{3x-1}{x\sqrt{x}}$. Show that the area bounded by the curve, the x-axis, and the lines $x = k$ and $x = 4k$, where k is a positive constant, is $\dfrac{6k-1}{\sqrt{k}}$.



1.37 — Question 46. The diagram below shows the curve with equation $y = \dfrac{x+1}{\sqrt{x}}$ and the line with equation $3x + 14y = 47$. Find the exact value of the shaded region.



1.37 — Question 47. Prerequisite knowledge for Q47: Graph Sketching: Radical Functions

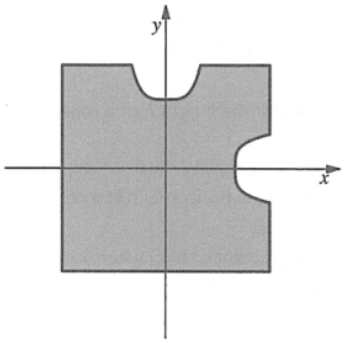
Find the exact finite area enclosed between the curve $y = \sqrt{x} - 16$, the x-axis and the y-axis.

1.37 — Question 48. Prerequisite knowledge for Q48: Graph Sketching: Cubic Graphs & Quartic Graphs

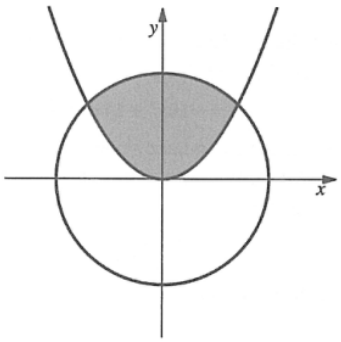
Find the exact finite area enclosed between the following curves and the x-axis:

a) $y = (x-1)(x-2)(x-3)$
b) $y = (x-1)^2(x-2)$
c) $y = (x-1)(x-2)^2$
d) $y = (x-1)^2(x-2)^2$

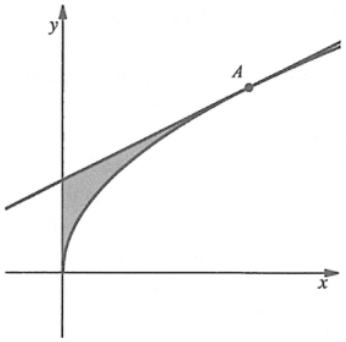
1.37 — Question 49. A puzzle piece is designed using a 6 cm by 6 cm square with the origin as its centre, as shown in the diagram below. Two identical curved shapes are cut out of the square to allow other pieces to be joined on the sides. The designer uses the equation $y = x^4 + 2$ to create one of the curved sections. Find the exact area of the puzzle piece.



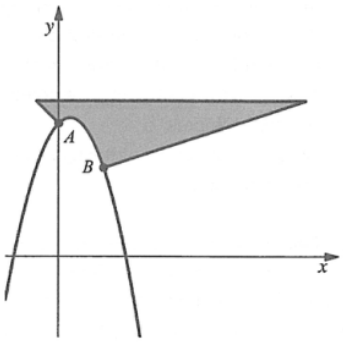
1.37 — Question 50. The diagram below shows the circle with equation $x^2 + y^2 = 2$ and the curve with equation $y = x^2$. Find the exact value of the shaded area.



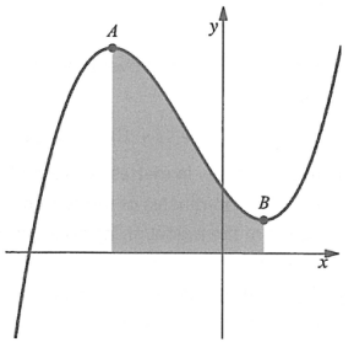
1.37 — Question 51. Prerequisite knowledge for Q51–Q53: Applications of Differentiation
The diagram below shows the curve with equation $y = 2\sqrt{x}$. The tangent to the curve at $x = 4$ is also shown in the diagram. The point A is where the tangent meets the curve. The shaded area is the region bounded by the curve, the tangent and the y-axis. Find the exact value of the shaded area.



1.37 — Question 52. The diagram below shows the curve with equation $y = (3 - x)(x + 2)$. The points A and B lie on the curve and have x-coordinates 0 and 2 respectively. The shaded region is formed by the curve, the line $y = 7$, the normal to the curve at A and the normal to the curve at B. Find the exact shaded area.



1.37 — Question 53. The diagram below shows the curve with equation $y = 2x^3 + 15x^2 - 144x + 500$. The points A and B lie on the turning points of the curve. Find the exact value of the shaded area.



1.38 — Question 1.

Applications of Differentiation

For each of the following situations, find the first differential, explain what it means in context and specify its units:

a) $C = 5.8x + 14.2$ where £C is the cost of the transport and x is the miles travelled.
b) $h = -t^2 + 6t + 135$ at $t = 10$ where h is the height of a ball (in metres) and t is the number of seconds passed.
c) $T = 0.01t^3 - 0.2t^2 + 20$ at $t = 5$ where T is the temperature (°C) of a substance and t is the number hours that have passed.

1.38 — Question 2. The spread of a virus over a period of 22 days may be modelled by the equation

$$N = -0.49t^2 + 10t + 3.54$$

where N is the number of people with the virus and t is the number of days passed. Find

a) the rate of spread of the virus on day 5
b) the rate of spread of the virus on day 15
c) the coordinate of the maximum of the curve of the model giving the points to 3 significant figures.
d) explain the meaning of the maximum point you found in part c)

1.38 — Question 3. The height h of a seedling in mm is given by

$$h = -0.028n^2 + 1.95n + 6.2$$

where n is the number of days passed.

a) Find $\frac{dh}{dn}$ and state its physical relevance.
b) Use this model to predict the day at which the seedling reaches maximum height and state its height.
c) How effective is the model in the long term?

1.38 — Question 4. The velocity of an object over t seconds may be given by

$$v = \frac{7}{3}t^3 - 8t^2 + 4t + 10, \qquad 0 \leq t \leq 3$$

where v is velocity in metres per second.

a) Find the coordinates of the stationary points of the model giving your answer to 3 significant figures.
b) Explain the meaning of stationary points in the context of the object.

1.38 — Question 5. The depth of water y cm below a model boat at time t seconds is modelled by

$$y = t^3 - 8t^2 + 15t + 48, \qquad 0 \leq t \leq 5$$

a) Find the greatest depth of water below the model boat.
b) Find the minimum value of $\frac{dy}{dt}$, the time at which it occurs and the depth of the water at this time.
c) State the vertical direction the boat is moving at the time found in part b)

1.38 — Question 6. A model for the height h of a roller coaster in feet is

$$h(x) = 2x^3 - 30x^2 + 94x + 150$$

where x is the horizontal distance in hundreds of feet. Find

a) the gradient of $h(x)$ when $x = 0$
b) the angle of inclination (in degrees) to the ground of the rollercoaster initially.
c) the highest point of the rollercoaster and where they occur, giving your answer to the nearest feet.

1.38 — Question 7.

Applications of Integration

The height h of a pile of sand above ground level can be modelled by

$$h(x) = -10x^3 + 14x^2 + \frac{3}{2}x$$

where x is the number of metres from one side. The mean height of the pile of sand may be given by

$$A = \frac{1}{2}\int_0^2 h(x)\,dx$$

Find the value of A

1.38 — Question 8. The value of some shares £V during the first 12 months of the year may be modelled by

$$V(t) = 0.5t^3 + 3.5t^2 - 12t + 50$$

where t is the number of months passed.

a) Use this model to predict the value of the shares when 12 months have passed.

The mean share of the shares value may be given by

$$A = \frac{1}{12}\int_0^{12} V(t)\,dt$$

b) Find the value of A